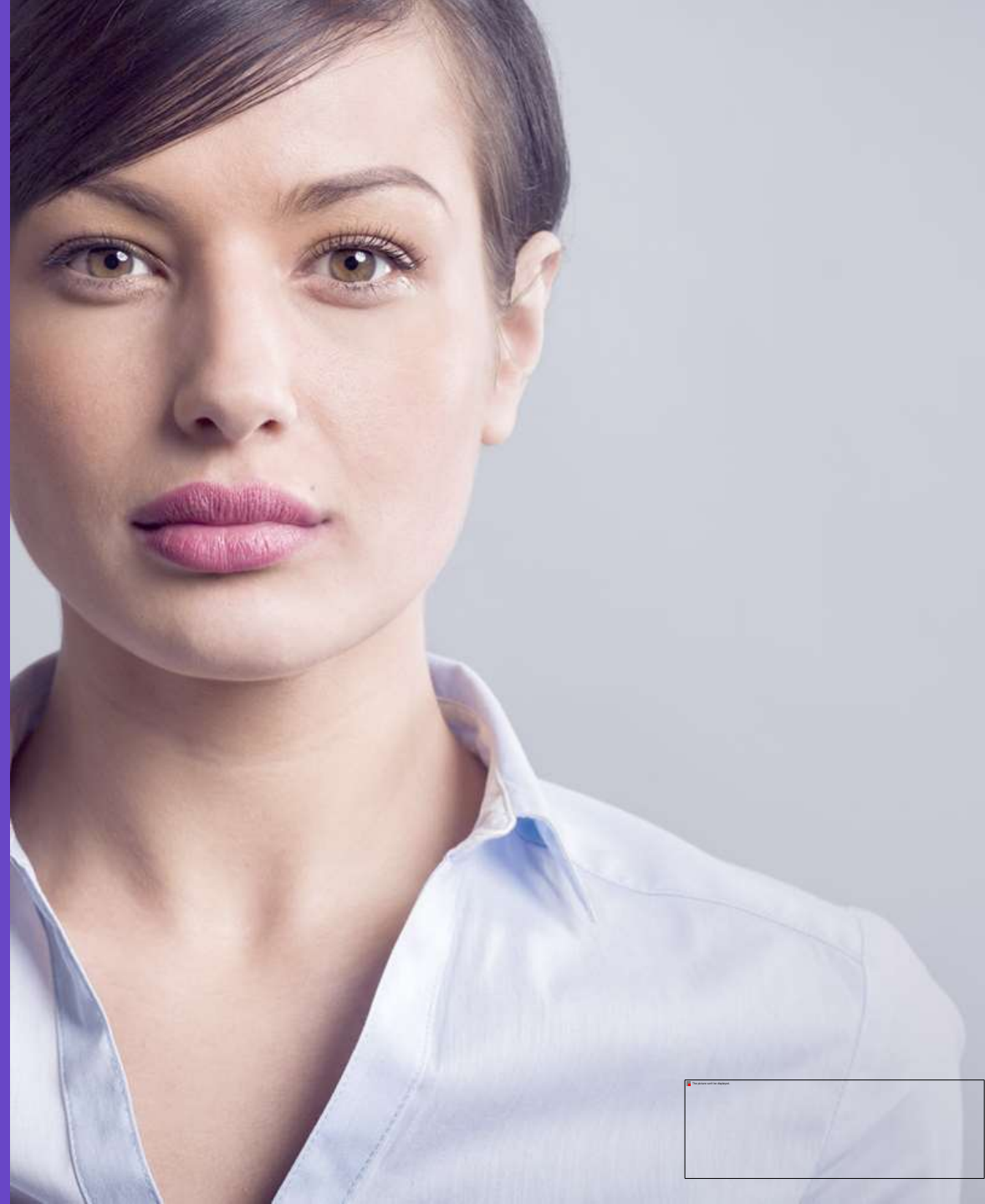


Drop Down List and its Methods



Contents

- Introduction to Select Class
- Different Select Class Methods
- How to deselect items
- How to select multiple items in Drop Down using Select Class Method
- How to Select Value & Handle Dropdown in Selenium without Select Class



Introduction

- In Selenium, the Select class provides the implementation of the HTML SELECT tag.
- A Select tag provides the helper methods with select and deselect options.
- As Select is an ordinary class, its object is created by the keyword New and also specifies the location of the web element.
- The Select is a Webdriver class that can work with the HTML SELECT tag. It exposes several “Select By” and “Deselect By” type methods. You can apply these methods to change the value in the dropdown and multiple select fields
- This class is a part of **org.openqa.selenium.support.ui.Select** Package
- **Select Class Syntax in Selenium:**

Select selectobj = new Select(<WebElement Object>);



Introduction

```
WebElement dropdown = driver.findElement(By.id("Browsers"));
Select select = new Select(dropdown);

// Alternatively, you can shorten the same code as given below.

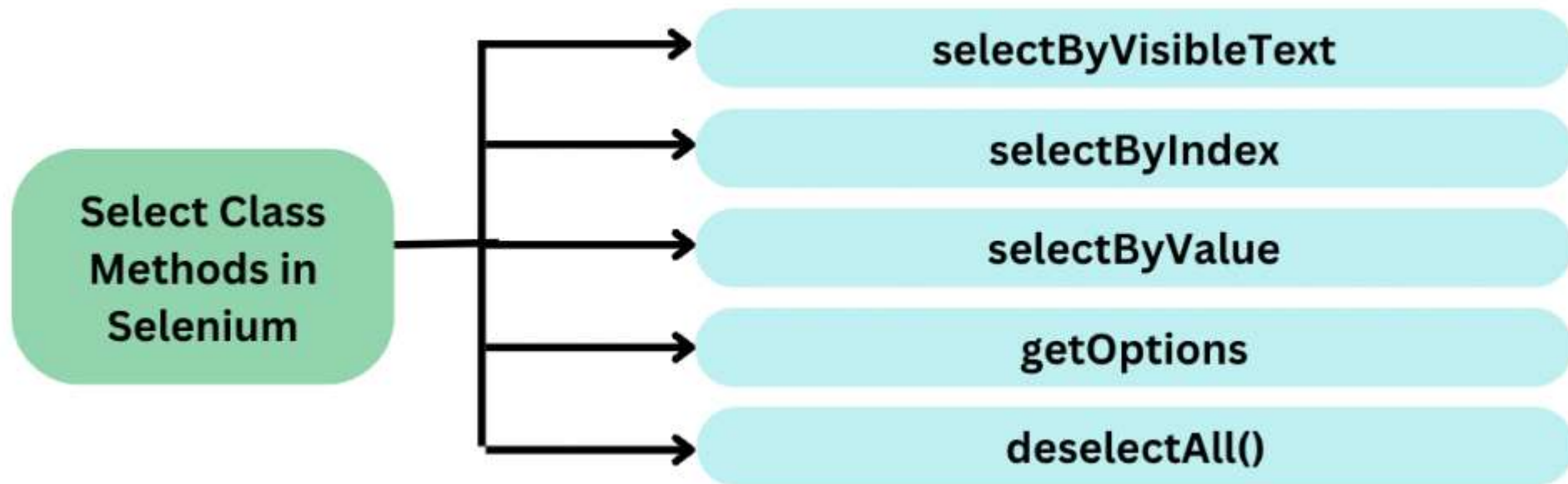
Select select = new Select(driver.findElement(By.id("Browsers")));
```



Different Select Class Methods

Different Select Class Methods

- The following are the most commonly used select class methods in selenium to deal with a drop-down list:



Different Select Class Methods

selectByVisibleText()

- This select class method is used to select one of the options in a drop-down box or an option among multiple selection boxes.
- It takes a parameter of String which is one of the values of Select element and it returns nothing.

Syntax

```
objSelect.selectByVisibleText("text");
```

Example:

```
Select objSelect =new Select(driver.findElement(By.id("search-box")));  
objSelect.selectByVisibleText("Automation");
```



Different Select Class Methods

selectByVisibleText()

1. selectByVisibleText()

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="UTF-8">
    <title>Select Example By Visible Text</title>
  </head>
  <body>
    <html>
      <body>
        <select name="Browsers">
          <option value="0" selected>Select...</option>
          <option value="1">Chrome</option>
          <option value="2">FireFox</option>
          <option value="3">IE</option>
          <option value="4">Safari</option>
          <option value="5">Opera</option>
        </select>
      </body>
    </html>
  </body>
</html>
```

Different Select Class Methods

selectByVisibleText()

1. selectByVisibleText()

```
WebElement dropdown = driver.findElement(By.name("Browser  
s"));

Select select = new Select(dropdown);
select.selectByVisibleText("FireFox");
```



Different Select Class Methods

selectByIndex()

- This method is similar to 'selectByVisibleText', but the difference here is that the user has to provide the index number for the option rather than text.
- It takes the integer parameter which is the index value of Select element and it returns nothing.

Syntax

```
objSelect.selectByIndex(int);
```

Example:

```
Select objSelect =new Select(driver.findElement(By.id("search-box")));  
  
objSelect.selectByIndex(3);
```



Different Select Class Methods

selectByIndex()

```
WebElement dropdown = driver.findElement(By.name("Browsers"));

Select select = new Select(dropdown);
select.selectByIndex(4);
```



Different Select Class Methods

selectByValue()

This method asks for the value of the desired option rather than the option text or an index. It takes a String parameter which is one of the values of Select element and it does not return anything.

Syntax

```
objSelect.selectByValue("text");
```

Example:

```
Select objSelect =new Select(driver.findElement(By.id("search-box")));  
objSelect.selectByValue("Auto");
```



Different Select Class Methods

selectByValue()

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="UTF-8">
    <title>Select Example By Value</title>
  </head>
  <body>
    <html>
      <body>
        <select name="Browsers">
          <option value="0" selected>Select...</option>
          <option value="Chrome">Chrome</option>
          <option value="FireFox">FireFox</option>
          <option value="IE">IE</option>
          <option value="Safari">Safari</option>
          <option value="Opera">Opera</option>
        </select>
      </body>
    </html>
  </body>
</html>
```

Different Select Class Methods

selectByValue()

```
WebElement dropdown = driver.findElement(By.name("Browsers"));

Select select = new Select(dropdown);
select.selectByValue("Opera");
```



Different Select Class Methods

getOptions()

- This method gets all the options belonging to the Select tag. It takes no parameter and returns List<WebElements>.

Syntax

```
objSelect.getOptions();
```

Example:

```
Select objSelect = new Select(driver.findElement(By.id("Search-box")));
```

```
List <WebElement> elementCount = objSelect.getOptions();
```

```
System.out.println(elementCount.size());
```



How to deselect items

DeSelect Methods

1. `deselectByIndex` Method : `objSelect.deselectByIndex(3);`

Use it when you want to deselect any option via its index. It accepts the index as an argument.

2. `deselectByValue` Method: `objSelect.deselectByValue("Opera");`

Use it when you wish to deselect all options using a value that matches the value in the dropdown options.

3. `deselectByVisibleText` Method : `objSelect.deselectByVisibleText("selenium");`

Use it to deselect all options by supplying the input text that matches the text in the dropdown options.



DeSelect Methods

4. deselectall Method: `objSelect.deselectAll();`

- Usage: Use it to turn off all the selected entries at once. But you can use this method if the `<Select>` tag has `<multiple>` attributes enabled in the HTML.
- Otherwise, it'll throw the `<NotImplemented>` exception. Hence, it's compulsory to have an attribute like `<multiple="multiple">` while defining the `<Select>` tag.



select multiple items

How to select multiple items in Drop Down using Select Class Method

- The multiple select attribute is a boolean expression. This method specifies that multiple options can be selected at once. These options vary for different operating systems and browsers.

isMultiple Method: `objSelect.isMultiple();`

`isMultiple()`: boolean – This method informs whether the Select element supports multiple selection options at the same time or not. This method accepts nothing and returns a boolean value (true/false).



Drop Down List and its Methods

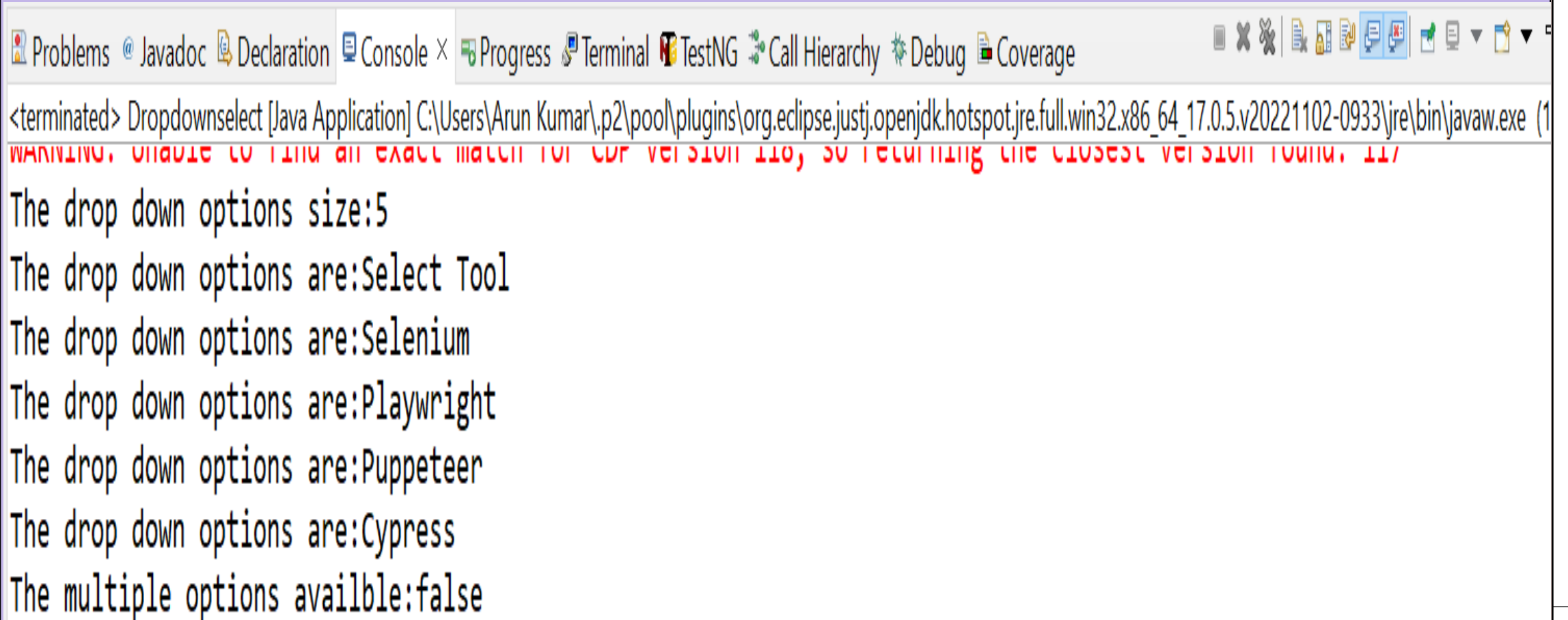
An Example Scenario

```
import org.openqa.selenium.support.ui.Select;
public class Dropdownselect {
    public static void main(String[] args) throws InterruptedException {
        // TODO Auto-generated method stub
        ChromeOptions options = new ChromeOptions();
        WebDriver driver = new ChromeDriver(options);
        driver.manage().window().maximize();
        driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));
        driver.get("https://leafground.com/select.xhtml");
        WebElement UIAutomation = driver.findElement(By.xpath("//select[@class='ui-selectonemenu']"));
        Select dropdown = new Select(UIAutomation);
        List<WebElement> list1 = dropdown.getOptions();
        System.out.println("The drop down options size:" +list1.size());
        for(WebElement ls : list1)
        {
            System.out.println("The drop down options are:" +ls.getText());
        }
        Boolean multiple = dropdown.isMultiple();
        System.out.println("The multiple options available:" +multiple);
        dropdown.selectByIndex(1); // by index
        dropdown.selectByVisibleText("Puppeteer"); //by visible text method
    }
}
```

Drop Down List and its Methods

An Example Scenario

Output:



```
<terminated> Dropdownselect [Java Application] C:\Users\Arun Kumar\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.5.v20221102-0933\jre\bin\javaw.exe (1
WARNING: Unable to find an exact match for CDP version 110, so returning the closest version found: 117
The drop down options size:5
The drop down options are:Select Tool
The drop down options are:Selenium
The drop down options are:Playwright
The drop down options are:Puppeteer
The drop down options are:Cypress
The multiple options availble:false
```

Drop Down List and its Methods

An Example Scenario

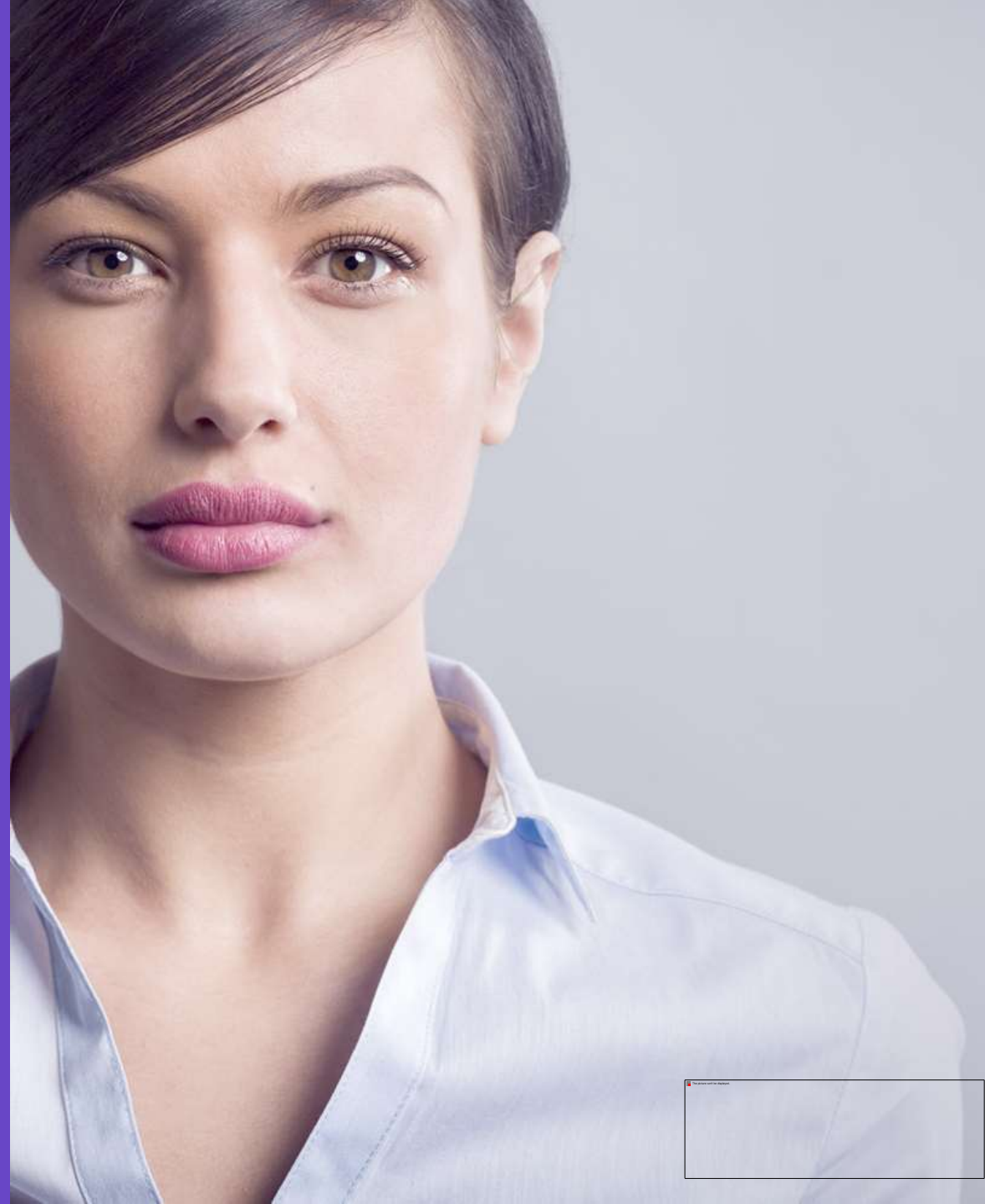
Output:

The screenshot displays a web application interface for Testleaf. The browser address bar shows the URL `leafground.com/select.xhtml`. A notification at the top states "Chrome is being controlled by automated test software." The application header includes the Testleaf logo and a search bar. A sidebar on the left contains navigation icons. The main content area is divided into two columns of dropdown menus:

- Left Column:**
 - "Which is your favorite UI Automation tool?" with a dropdown menu showing "Puppeteer".
 - "Choose your preferred country." with a dropdown menu showing "Select Country".
 - "Confirm Cities belongs to Country is loaded" with a dropdown menu showing "Select City".
- Right Column:**
 - "Choose the Course" with a dropdown menu showing "Choose Course".
 - "Choose language randomly" with a dropdown menu showing "Select Language" and a settings gear icon.
 - "Select 'Two' irrespective of the language chosen" with a dropdown menu showing "Select Values".

The footer of the application displays "2022 - All Rights Reserved".

How to Select Value & Handle Dropdown in Selenium without Select Class



Introduction

- Drop downs in a website could be created in several different ways.
- Some dropdowns are created using `<select>` HTML tag and some others are created using `<ul>` ,`<li>`, `<button>` and `<div>` tags.
- Some dropdowns are dynamic in nature which means after clicking or selecting any option, the drop downs values would populate accordingly.
- And in some cases, you need to [mouse hover on an element in Selenium](#) to see the drop-down options.
-



Introduction

- [Selenium WebDriver](#) provides Select class which can be used only for drop down created using <select> HTML tag.
- Select class has methods such as selectByVisibleText(), selectByIndex() and selectByValue() to select the desired option from dropdown.
- However, for NonSelect dropdowns, Select class cannot be used.



How to Select Dropdown in Selenium?

You can handle dropdown in Selenium using Select Class and without using Select Class. Below are 5 different methods that can be used to select value in dropdown in Selenium without using Select Class. These methods are:

- 1.By storing all the options in List and iterating through it
- 2.By creating Custom Locator and without iterating the List
- 3.By using [JavaScriptExecutor](#) class
- 4.By using [sendKeys](#)
- 5.By using [Actions Class](#)



How to Select Dropdown in Selenium?

1.By storing all the options in List and iterating through it

- You can handle Dropdown in Selenium by storing all the options in the form of a List and then iterate through it



How to Select Value & Handle Dropdown in Selenium without Select Class

An Example Scenario

```
public class ListDemo {  
    public static void main(String[] args) {  
        ChromeOptions options = new ChromeOptions();  
        WebDriver driver = new ChromeDriver(options);  
        driver.manage().window().maximize();  
        driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));  
        driver.get("https://www.bstackdemo.com/");  
        driver.findElement(By.xpath("//select")).click();  
        //Method 1: By storing all the options in List and iterating through it  
        List<WebElement> allOptions = driver.findElements(By.cssSelector("select option"));  
        System.out.println("Options Size:" + allOptions.size());  
        String option = "Lowest to highest";  
        for (int i = 0; i < allOptions.size(); i++) {  
            if (allOptions.get(i).getText().contains(option)) {  
                allOptions.get(i).click();  
                System.out.println("clicked");  
                break;  
            }  
        }  
    }  
}
```

How to Select Dropdown in Selenium?

2. By creating Custom Locator and without iterating the List:

- In this method, you have to create custom [Locator in Selenium](#), and then handle the drop down without iterating the List

//Method 2: By creating Custom Locator and without iterating the List

```
String option1="Highest to lowest";
```

```
WebElement dropdown =driver.findElement(By.xpath("//select/option[contains(text(),  
'" +option1+"' )]"));
```

```
dropdown.click();
```



How to Select Dropdown in Selenium?

3. By using [JavaScriptExecutor](#) class:

- [JavaScriptExecutor](#) is an Interface that helps to execute JavaScript through Selenium WebDriver.
- You can use the executeScript method of JavaScriptExecutor interface to select an option from the drop down by using the value property of the element

//Method 3: By using JavaScriptExecutor class

```
WebElement option2 = driver.findElement(By.xpath("//select"));
JavascriptExecutor js = (JavascriptExecutor) driver;
js.executeScript("arguments[0].value='lowestprice'", option2);
```



How to Select Dropdown in Selenium?

4. By using sendKeys

- You can use sendKeys() method to select an option from a drop down by passing the element's value as sendKeys char sequence

//Method 4: By using sendKeys method

```
driver.findElement(By.xpath("//select")).sendKeys("highestprice");
```



How to Select Dropdown in Selenium?

5. By using Actions Class

- Some drop-downs need to be hovered to display all the options present in it and then click on it.
- [Actions class of Selenium](#) is used for handling keyboard and mouse events

//Method 5: By using Actions Class

```
WebElement ele = driver.findElement(By.xpath("//select"));
Actions action=new Actions(driver);
action.moveToElement(ele).perform();
WebElement customOption=driver.findElement(By.xpath("//option[text()='Lowest to highest']"));
customOption.click();
```



How to Select Value & Handle Dropdown in Selenium without Select Class



```
1 | <ol>

1 | <html>
2 |
3 | <head>
4 |   <title>Numbered List Example</title>
5 | </head>
6 |
7 | <body>
8 |   <h2>Ordered List with Numbers</h2>
9 |   <ol>
10 |     <li>JavaScript</li>
11 |     <li>Python</li>
12 |     <li>Java</li>
13 |     <li>C++</li>
14 |     <li>C#</li>
15 |   </ol>
16 | </body>
17 |
18 | </html>
```

How to Select Dropdown in Selenium?

```
//Locate the ordered list using its tag name
WebElement orderedList = driver.findElement(By.tagName(("ol")));

//Fetch all the list items
List<WebElement> listItems = orderedList.findElements(By.tagName("li"));

//Iterate through the list and print the contents
for (int i = 0; i < listItems.size(); i++) {
    System.out.println(listItems.get(i).getText());
}

//Close the main window
driver.quit();
```



How to Select Value & Handle Dropdown in Selenium without Select Class

How to Select Dropdown in Selenium?

```
WARNING: Unable to find an exact match for CDP v
JavaScript
Python
Java
C++
C#
```



Sorting dropdown list

```
WebElement dropdownElement = driver.findElement(By.id("country"));

Select dropdown = new Select(dropdownElement);

List<WebElement> options = dropdown.getOptions();

    // Extract the text of the options and store them in a list
    List<String> optionTextList = new ArrayList<>();
    for (WebElement option : options) {
        optionTextList.add(option.getText());
    }

    // Sort the list of option texts
    Collections.sort(optionTextList);
```

Auto Suggestions List

Handling Auto Suggestions

- Find multiple elements and store them in a List.

```
WebElement searchBox = driver.findElement(By.id("search"));
searchBox.sendKeys("selenium");

WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));

List<WebElement> suggestions = wait.until(
    ExpectedConditions.visibilityOfAllElementsLocatedBy(By.xpath("//ul/li")));

for (WebElement option : suggestions) {
    if (option.getText().equals("selenium webdriver")) {
        option.click();
        break;
    }
}
```